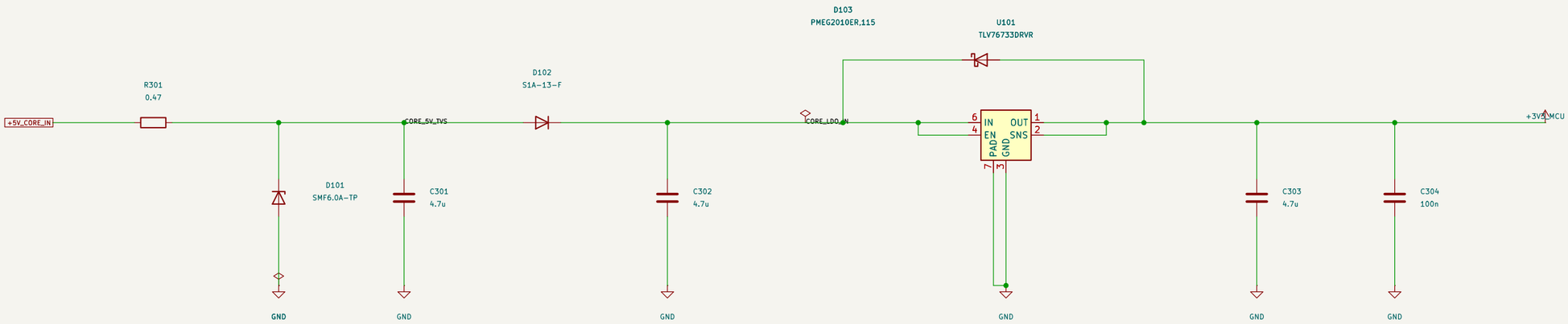
5 V INPUT / INDEPENDENT 3.3 V MCU RAIL

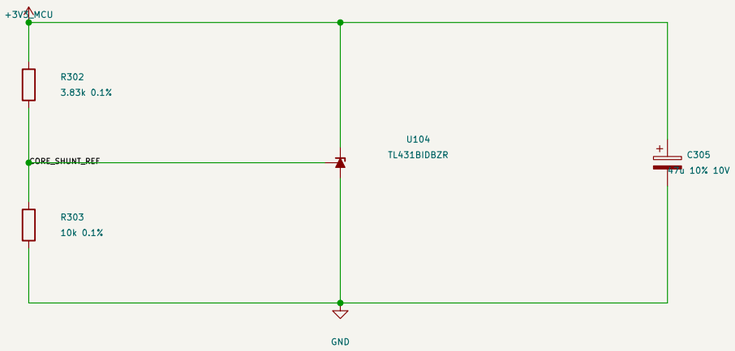
Regulated baseboard supply: +5V_CORE_IN = 4.75–5.25 V at header. Shared GND; no galvanic isolation.
100 mA continuous core budget; optional CAN/sensors require a fresh load and thermal budget.



D102 prevents output/injection current returning to the 5 V header.
D103 bypasses LDO reverse stress when OUT remains charged.
TVS clamp is 10.3 V at its specified pulse; U101 accepts up to 16 V.
No sustained 24/60 V miswire or IEC surge qualification is claimed.

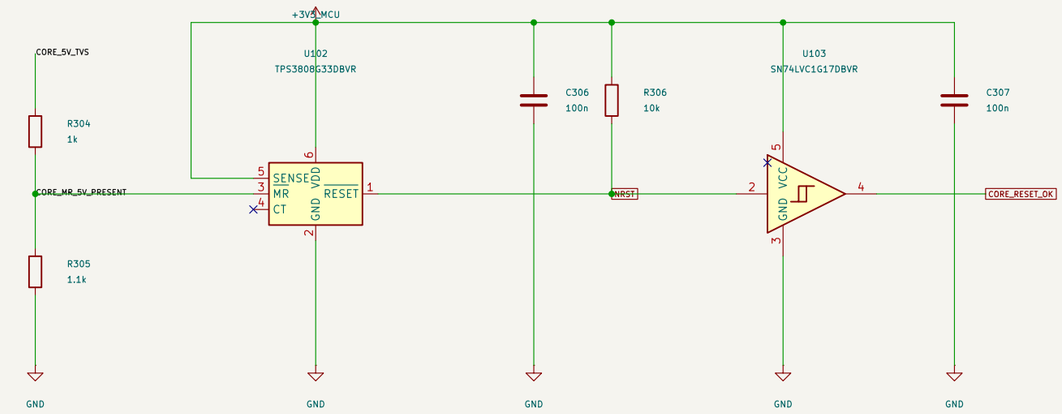
U101 SNS (pin 2) connects to OUT. GND pins 3/5 and EP7 to GND.
Input/output capacitors and EP thermal copper stay on the core.
The 1 A IC rating does not increase the header/core 100 mA budget.

ADC CLAMP–CURRENT ABSORBER / LOCAL 3.3 V BULK



Nominal clamp: 2.495 x (1 + 3.83k/10k) + IREF x 3.83k ~ 3.46 V.
Positive transient injection budget <=50 mA: no hard current limit.
8 ADC paths: 360R; BASE_READY / FAULT_N: 1k each.
Inputs 5.25 V, rail >=3.25 V, zero diode Vf, resistors −1%: total 48.93 mA.
The downstream 100R does not limit rail injection.
ADC RC: (360+100)R x 1nF = 0.46 us; add external source impedance.
C305 required for shunt stability; validate startup/pulse/ringing on prototype.
No sustained 60 V fault immunity is claimed.

LOCAL RESET / BUFFERED ACTIVE–HIGH PERMISSION



RESET is open drain; CT open gives nominal 20 ms release delay.
MR divider requires 5 V presence even if ADC clamps back–power 3.3 V.
RESET and CB1 stay on the core support page; SWD is on the base.
CORE_RESET_OK goes to the separate RUN_ARM gate.
PC7/RUN_ARM is required before CTRL_READY can be HIGH.

CONTROLLER | local regulated 5 V input | independent 3.3 V | common ground
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

Sheet: /Local power and reset/
File: core_supply.kicad_sch

Title: Minimum core — independent power and local reset

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 5/7